higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MECHANICAL DRAWING AND DESIGN N5

27 MARCH 2019

This marking guideline consists of 10 pages.

Copyright reserved

QUESTION 1

Data:

$l_1 = 100 \ \ mm\,;\ \ t_1 = 12 \ \ mm\,;\ \ \sigma_t = 80 \ \ MPa\,;\tau = 60 \ \ MPa\,;\ P = 100 \ \ kN\,;\ \ l_2 = ?$

Solution:

The length l_2 of each parallel fillet:

- $t = \frac{1}{\sqrt{2}} t_1 = \frac{1}{\sqrt{2}}(12) = 8{,}485 \ \ mm$
- $A_1 = l_1 t = (100)(8{,}485) = 848{,}528 \ \ mm^2$

$\therefore \ \ P_1 = \sigma_t \ \ A_1 = (80)(848{,}528) = 67{,}882 \ \ kN$

- $A_2 = 2(l_2 - 2t)t = 2\left[l_2 - 2(8{,}485)\right](8{,}485) = 16{,}971\, l_2 - 288 \ \ mm^2$

$\therefore \ \ P_2 = \tau \ \ A_2 = (60)(16{,}971\, l_2 - 288) = 1 \ \ 018{,}234\, l_2 - 17 \ \ 280 \ \ N$

But:

- $P = P_1 + P_2$

$\therefore 100 \times 10^3 = 67{,}882 \times 10^3 + (1 \ \ 018{,}234\, l_2 - 17 \ \ 280)$

$\therefore l_2 = 48{,}513 \ \ mm$ **[9]**

Copyright reserved

QUESTION 2

Data:

$D_s = 50\ \ mm;\ \ d_h = 55\ \ mm;\ \ G_h = G_s;\ \ L_h = L_s;\ \ \theta_h = \theta_s$

Solution:

2.1 The outside diameter of the sleeve:

Given: $T_h = 2{,}5\,T_s$

$\bullet\ \theta_h = \theta_s$

$$\frac{10{,}2\,T_h\,L}{G\left(D_h^{\,4} - d_h^{\,4}\right)} = \frac{10{,}2\,T_s\,L}{G\,D_s^{\,4}}$$

$$\frac{2{,}5\,T_s}{D_h^{\,4} - d_h^{\,4}} = \frac{T_s}{D_s^{\,4}}$$

$$\frac{2{,}5}{D_h^{\,4} - (55)^4} = \frac{1}{(50)^4}$$

$$D_h^{\,4} - (55)^4 = 2{,}5 \times (50)^4$$

$$\therefore D_h = \sqrt[4]{2{,}5 \times (50)^4 + (55)^4} = 70{,}\ 551\ \ mm$$

$$\therefore D_h = 71\ \ mm$$

NOTE: The upper value for the outside diameter is preferred to the lower value as it favours the resistance of the hollow shaft to torsion. (6)

2.2 The torque transmitted by the compound shaft:

Given: $\tau = 80\ \ MPa$

$$\bullet\ T_s = \frac{\pi}{16} D_s^{\,3} \tau = \frac{\pi}{16}(0{,}05)^3 \times \left(80 \times 10^6\right) = 1\ \ 963{,}945\ \ Nm$$

$$\bullet\ T_h = \frac{\pi}{16} \frac{D_h^{\,4} - d_h^{\,4}}{D_h}\ \ \tau = \frac{\pi}{16} \times \frac{(0{,}071)^4 - (0{,}055)^4}{0{,}071} \times \left(80 \times 10^6\right)$$

$$\therefore T_h = 3\ \ 597{,}578\ \ Nm$$

Therefore:

$$T_t = T_s + T_s = 1\ \ 963{,}945 + 3\ \ 597{,}578 = 5\ \ 561{,}523\ \ Nm$$ (5)

Copyright reserved

2.3 The power transmitted:

Given: $N = 900 \ \ r/sec$; $1.x = 1{,}125$

- $T_{mean} = \dfrac{T_{max}}{1.x} = \dfrac{5\ \ 561{,}523}{1{,}125} = 4\ \ 943{,}576 \ \ Nm$

- $P = \dfrac{2\pi N\, T_{mean}}{60} = \dfrac{2\pi(900)(4\ \ 943{,}576)}{60} = 465\ \ 921{,}061 \ \ W = 465{,}921 \ \ kW$

- $T_h = \dfrac{\pi}{16}\dfrac{D_h{}^4 - \ d_h{}^4}{D_h} \ \ \tau = \dfrac{\pi}{16} \times \dfrac{(0{,}071)^4 - (0{,}055)^4}{0{,}071} \times (80 \times 10^6)$

$\therefore T_h = 3\ \ 597{,}578 \ \ Nm$

Therefore:

$T_t = T_s + T_s = 1\ \ 963{,}945 + 3\ \ 597{,}578 = 5\ \ 561{,}523 \ \ Nm$ (4)

[15]

QUESTION 3

3.1 Data:

$n = 4$; $d = 60 \ \ mm$; $D = 150 \ \ mm$; $A_c = \dfrac{3}{5}A_s$; $\tau = 5 \ \ MPa$; $\sigma_c = 12 \ \ MPa$

Solution:

3.1.1 The minimum depth of the claws:

- $A_s = \dfrac{\pi}{8}(D^2 - d^2) = \dfrac{\pi}{8}(0{,}15^2 - 0{,}06^2) = 7{,}422 \times 10^{-3} \ \ m^2$

- $A_c = \dfrac{3}{5}A_s = \dfrac{3}{5}(7{,}422 \times 10^{-3}\ ) = 4{,}453 \times 10^{-3} \ \ m^2$

But:

- $A_c = nw \times \dfrac{1}{2}(D - d)$

$\therefore 4{,}453 \times 10^{-3} = (4)w \times \dfrac{1}{2}(0{,}15 - 0{,}06)$

$w = 24{,}74 \ \ mm$

$\therefore w = 25 \ \ mm$ (5)

3.1.2 The maximum power transmitted by the claws:

Data:

$\omega = 80\pi \ \ rad/s$; $1.x = 1{,}15$

Solution:

$$\bullet\ R_m = \frac{D+d}{4} = \frac{150+60}{4} = 52{,}5 \ \ mm$$

Crushing:

$$\bullet\ T_{max} = n\left[\frac{1}{2}(D-d)\,w\,\sigma_c\right]R_m$$

$$T_{max} = (4)\left[\frac{1}{2}(0{,}15-0{,}06)(0{,}025)\left(12\times10^{6}\right)\right]\left(52{,}5\times10^{-3}\right)$$

$$\therefore T_{max} = 2\ \ 835 \ \ Nm$$

Shearing:

$$\bullet\ T_{max} = \left[\frac{1}{2}\times\frac{\pi}{4}\left(D^2-d^2\right)\tau\right]R_m$$

$$T_{max} = \left[\frac{1}{2}\times\frac{\pi}{4}\left(0{,}15^2-0{,}06^2\right)\left(5\times10^{6}\right)\right]\left(52{,}5\times10^{-3}\right)$$

$$\therefore T_{max} = 1\ \ 948{,}278 \ \ Nm$$

Therefore, use $T_{max} = 1\ \ 948{,}278 \ \ Nm$

$$\bullet\ T_{mean} = \frac{T_{max}}{1.x} = \frac{1\ \ 948{,}278}{1{,}15} = 1\ \ 694{,}155 \ \ Nm$$

$$\bullet\ P = T_{mean}\times\omega = (1\ \ 694{,}155)(80\pi) = 425{,}788 \ \ kW$$ (10)

3.2 Data:

$D_s = 30 \ \ mm ; \tau = \frac{75}{2,5} \ \ MPa = 30 \ \ MPa$

Solution:

3.2.1 The maximum length of the lever:

Given: $F = 250 \ \ N$

$\bullet \ T_{max} = F \times l = 250\,l$

and

$\bullet \ T_{max} = \frac{\pi}{16} D_s{}^3 \tau = \frac{\pi}{16}(0,03)^3\left(30 \times 10^6\right) = 159,043 \ \ Nm$

Therefore:

$250\,l = 159,043$
$\therefore l = 636,172 \ \ mm$ (4)

3.2.2 The average diameter of a taper pin:

$\bullet \ T_{max} = \frac{\pi}{4} d_p{}^2 \tau \, D_s$

$159,043 \ = \frac{\pi}{4} d_p{}^2 (30 \times 10^6)(0,03)$

$\therefore d_p = 15 \ \ mm$ (2)

[21]

QUESTION 4

Data:
$D = 450 \ \ mm ; sl = 360 \ \ mm ; p = 1,45 \ \ MPa$

Solution:

4.1 The force on the piston:

$F = \frac{\pi}{4} D^2 p = \frac{\pi}{4}(0,45)^2\left(1,45 \times 10^6\right) = 230,613 \ \ kN$ (2)

4.2 The magnitude of the force on the crank pin:

Given: $L = 750 \ \ mm$

[**NOTE:** The crank radius is at right angles with the piston rod centre line]

$$\bullet \sin\alpha = \frac{r}{L} = \frac{\frac{1}{2}sl}{L} = \frac{\frac{1}{2}(360)}{750}$$
$$\therefore \alpha = 13{,}887\,°$$

$$\bullet \cos\alpha = \frac{F}{k}$$
$$\cos 13{,}887\,° = \frac{230{,}613}{k}$$
$$\therefore k = 237{,}557 \ \ kN$$ (5)

4.3 The pressure between the crank pin and the connecting rod:

Given: $L = 150 \ \ mm \,;\, D = 125 \ \ mm$

$$p = \frac{k}{A} = \frac{k}{DL} = \frac{237{,}557 \times 10^3}{(0{,}125)(0{,}15)} = 12{,}67 \ \ MPa$$ (3)

4.4 The length and the width of the crosshead shoes:

Given: $p = 7500 \ \ kPa \,;\, l = 2{,}5 \ \ w$

$$\bullet \tan\alpha = \frac{R}{F}$$
$$\tan 13{,}887\,° = \frac{R}{230{,}613}$$
$$\therefore R = 57{,}015 \ \ kN$$

$$\bullet A = lw = (2{,}5 \ \ w)w = 2{,}5 \ \ w^2$$

$$\bullet p = \frac{R}{A}$$
$$p = \frac{R}{2{,}5\,w^2}$$
$$7 \ \ 500 \times 10^3 = \frac{57{,}015 \times 10^3}{2{,}5 \ \ w^2}$$

$$\therefore w^2 = 3{,}0408 \times 10^{-3}$$
$$\therefore w = 55{,}143 \ \ mm$$

Say $w = 56 \ \ mm$ and $l = 2{,}5(56) = 140 \ \ mm$ (7)
[17]

QUESTION 5

5.1 Data:

$P = 40\ \ kW\ ;\ \ D_s = 50\ \ mm\ ;\ \ N = 9\ \ r/sec\ ;\ D = d = 250\ \ mm$

$C = 900\ \ mm\ ;\ \ t = 15\ \ mm\ ;\ \ T_2 = 0{,}4\ \ T_1$

Solution:

Crossed flat-belt drive

5.1.1 The total length of the belt:

$$L = \frac{\pi}{2}(D + d + 2t) + \frac{(D + d + 2t)^2}{4C} + 2C$$

$$L = \frac{\pi}{2}[250 + 250 + 2(15)] + \frac{[250 + 250 + 2(15)]^2}{4(900)} + 2(900)$$

$$\therefore L = 2\ \ 710{,}55\ \ mm$$ (2)

5.1.2 The coefficient of friction between the belt and the pulleys:

Since: $D = d$, $\beta = 0$ and $\theta_1 = \theta_2 = 180° = \pi\ \ rad$

Therefore:

$$\bullet \sin\beta = \frac{D + d}{2C} = \frac{250 + 250}{2(900)}$$

$$\therefore \beta = 16{,}128°$$

$$\bullet \theta = 180° + 2\beta = 180° + 2(16{,}128°) = 212{,}255° = 3{,}705\ \ rad$$

$$\bullet \frac{T_1}{T_2} = e^{\mu\theta}$$

$$\frac{T_1}{0{,}4\,T_1} = e^{\mu(3.705)}$$

$$\mu = \frac{1}{3{,}705}\ln\frac{1}{0{,}4}$$

$$\therefore \mu = 0{,}247$$ (8)

5.1.3 The tight- and slack-side tensions:

Given: $\Delta P = 0{,}1\%\ \ of\ \ P$ (therefore: $\eta = 99{,}9\%$)

$$\bullet v = \pi N(D + t) = \pi(9)(0{,}25 + 0{,}015) = 7{,}493\ \ m/s$$

$$\bullet \eta P = (T_1 - T_2)v$$

$$(0{,}999)(40 \times 10^3) = (T_1 - 0{,}4\,T_1)(7{,}493)$$

$$\therefore T_1 = 8\ \ 888{,}296\ \ N \text{ and } T_2 = 0{,}4(8\ \ 888{,}296) = 3\ \ 555{,}318\ \ N$$ (8)

Copyright reserved

5.1.4 The coefficient of friction between the driven shaft and the bearings:

Given: $P_{\mu}=10\ \ W$

- $v_s=\pi ND=\pi(9)\left(50\times10^{-3}\right)=1{,}414\ \ m/s$
- $N_R=T_1+T_2=8\ \ 888{,}296+3\ \ 555{,}318=12\ \ 443{,}614\ \ N$
- $P_{\mu}=F_{\mu}v_s$

$P_{\mu}=(\mu N_R)v_s$
$10=(\mu\times12\ \ 443{,}614)(1{,}414)$
$\therefore\mu=0{,}0006\ \left[\text{or}\ \ 0{,}568\times10^{-3}\right]$ (6)

5.2 The maximum bearing stress exerted on the bearings:

Given: $L=75\ \ mm$

- $F=T_1+T_2=12\ \ 443{,}614\ \ N$

(from QUESTION 5.1.4 above)
Assuming upward reactions:

- $\circlearrowleft M_B=\circlearrowleft M_B$

$T\times0{,}2=A\times1$
$12\ \ 443{,}614\times0{,}2=A$
$\therefore A=2\ \ 488{,}723\ \ N$

- $\circlearrowleft M_A=\circlearrowleft M_A$

$B\times1+T\times1{,}2=0$
$B\times1+12\ \ 443{,}614\times1{,}2=0$
$\therefore B=-14\ \ 932{,}337\ \ N$
$\therefore B=14\ \ 932{,}337\ \ N\ (downward)$

- $p=\dfrac{B}{DL}=\dfrac{14\ \ 932{,}337}{(0{,}05)(0{,}075)}$

$\therefore p=3{,}982\ \ MPa$

(5)

OR

- $F=T_1+T_2=12\ \ 443{,}614\ \ N$

(from QUESTION 5.1.4 above)
Assuming upward reactions:

- $\circlearrowleft M_A=\circlearrowleft M_A$

$B\times1+T\times1{,}2=0$
$B\times1+12\ \ 443{,}614\times1{,}2=0$
$\therefore B=-14\ \ 932{,}337\ \ N$
$\therefore B=14\ \ 932{,}337\ \ N(downward)$

- $p=\dfrac{B}{DL}=\dfrac{14\ \ 932{,}337}{(0{,}05)(0{,}075)}$

$\therefore p=3{,}982\ \ MPa$

(5)
[29]

QUESTION 6

Data:

$\frac{D}{t}=62,5\ ;\ p=5\ \ MPa$

Solution:

6.1 The circumferential stress:

- $\sigma_c=\frac{p}{2}\left(\frac{D}{t}\right)=\frac{5}{2}(62,5)=156,25\ \ MPa$

- $\sigma_l=\frac{p}{4}\left(\frac{D}{t}\right)=\frac{5}{4}(62,5)=78,125\ \ MPa$

$\left[Award\ \ 1\ \ mark\ \ ONCE\ \ for\ \ the\ \ substitution\ \ \frac{D}{t}=62,5\right]$ (3)

6.2 The axially applied compressive load:

Given: $t=4\ \ mm$

- $D=62,5\,t=62,5(4)=250\ \ mm$

- $A=\pi Dt=\pi(0,25)(0,004)=3,142\times10^{-3}\ \ m^2$

- $F=\sigma_l A=\left(78,125\times10^{6}\right)\left(3,142\times10^{-3}\right)=245,437\ \ kN$ (6)

[9]

TOTAL: 100

Copyright reserved